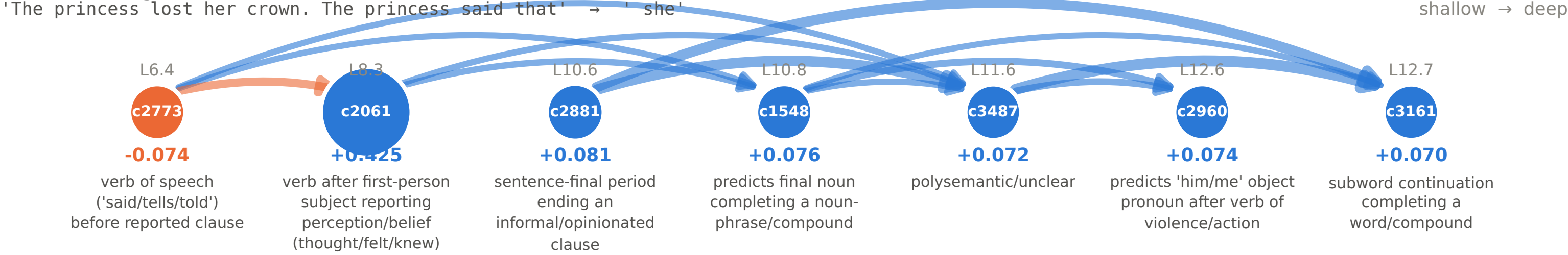


# Attribution graphs over parameter components — VPD §5

Node size is the direct effect on  $\log p(\text{target})$  when that component's owned mass is scaled by 4. An arc  $c' \rightarrow c$  is  $DE(c) - DE(c \mid c' \text{ already perturbed})$ : how much of  $c$ 's effect depends on  $c'$  having run first. Order is by depth, the mass-weighted mean layer of a component's parameters. VPD ABLATES here; that fails on this decomposition — a geometric partition of trained weights, not something optimised for ablation sufficiency. Zeroing a whole component moves  $\log p$  by  $\sim 0.002$  nats; scaling it by 4 moves it by up to 5.

## Pronoun agreement — a distributed circuit



## Copying a novel string — a star

